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Evaluating Suppression Index as a Cognitive Marker: Insights from Visual Search Tasks for Alzheimer's Assessment and Intervention

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Abstract: Alzheimer's Disease (AD) is marked by cognitive decline, including deficits in visual search and spatial suppression. While endogenous cues (e.g., visual words and audio) influence attention, their role in modulating goal-directed spatial suppression during visual search tasks remains underexplored in AD. The idea behind this study is to use the suppression index to look into how changing internal cues affect goal-directed spatial suppression and how these cues relate to cognitive tests like the Mini-Mental State Examination (MMSE) in Alzheimer's disease. This framework is integrated with spatial filtering during spatial suppression tasks. An attention mechanism simulates how cues modulate attention. This study is based on visual search data (reaction times, accuracy) and dynamic word/audio cues, with a regression module using MMSE scores. This work aims to uncover cognitive mechanisms of spatial suppression under dynamic cueing, provide a view for early AD diagnosis by linking visual search performance to MMSE scores, and explore personalized cognitive interventions using dynamic cues. This paper investigates the Suppression Index as a possible indicator of cognitive ability in Alzheimer's sufferers. The results point to similar cognitive processes since the suppression patterns show a significant overlap. These results highlight the potential of the experiments described in this work as diagnostic and therapeutic tools for early Alzheimer's disease diagnosis and cognitive training.

Keywords: Spatial Suppression , Cognitive Decline, Suppression index , Mini-Mental State Examination (MMSE)

1. Introduction

Let us imagine ourselves and a family member engaged in a jigsaw puzzle project. The puzzle contains various shapes and colors, and the goal is to find and fit the correct pieces together. For instance, you might be looking for a piece with a specific shape, such as a straight edge or a corner piece, to complete the border of the puzzle. We scan the puzzle pieces, focusing on the shape and color of the ones you need. Spatial suppression helps us ignore irrelevant pieces (e.g., those with the wrong shape or color) and focus on pieces that match the target. Once we find a potential piece, we can confirm whether it fits by examining its shape and trying to place it in

the puzzle. The process is relatively quick and efficient because our brain can filter out distractions and prioritize relevant visual information [1]. AD patients often struggle with visual processing and attention, which makes it challenging to distinguish the target piece from the clutter. The cluttered environment of puzzle pieces overwhelms their visual system, as they have difficulty suppressing irrelevant stimuli (e.g., pieces with the wrong shape or color). They may fixate on distractions such as bright colors or interesting patterns and lose track of their goal—finding the correct piece [2]. AD patients may experience visual crowding, where the presence of many pieces in close proximity makes it harder to distinguish individual shapes. They might also struggle with working memory, forgetting what the target piece looks like or why they are searching for it in the first place. Failing to find the correct piece can lead to frustration, confusion, or anxiety, further impairing their cognitive ability to focus on the task [3].

2. Related Work

Alzheimer's Disease (AD) is characterized by progressive neurodegeneration, marked by cortical disconnection and atrophy in temporo-parietal regions, which disrupts the integration of sensory and attentional processes [4, 8]. These changes impair controlled information processing, leading to deficits in visual search efficiency, spatial suppression, and inhibitory control [5]. For instance, individuals with AD exhibit delayed target detection and prolonged fixation durations during visual search tasks, reflecting inefficient attentional strategies [3, 7]. Eye-tracking studies further reveal that AD patients produce shorter saccades during reading and show visuomotor coordination deficits, underscoring the degradation of frontoparietal networks critical for spatial attention [7, 9]. Such impairments highlight the need for a cognitive marker sensitive to preclinical changes, as traditional tools like the Mini-Mental State Examination (MMSE) often fail to detect subtle early-stage deficits [11].

The MMSE, while widely used to assess cognitive domains such as orientation, memory, and visuospatial skills, lacks granularity in capturing dynamic attentional disruptions [11]. To address this gap, recent research has focused on spatial suppression—a mechanism by which the brain prioritizes task-relevant stimuli while suppressing distractors [12, 15]. In AD, impaired inhibitory control disrupts this process, resulting in heightened interference from high-probability distractors and slower target processing [13]. For example, Van Moorselaar and Theeuwes [15] demonstrated that statistical regularities in distractor locations enhance suppression efficiency in healthy adults, but this adaptability is diminished in AD due to degraded experience-dependent plasticity. These findings align with fMRI evidence showing reduced activation in parietal and frontal regions, which are pivotal for top-down attentional control [8, 17].

Visual search paradigms, such as the additional singleton task, have been instrumental in dissecting attentional selection from response processes [16]. However, compound search tasks, which always include a target, may conflate these mechanisms. Visual detection tasks (e.g., present/absent judgements) offer a more precise framework for studying suppression, as they eliminate target-generated priority signals [15]. Wang et al. [18] found that participants ignored high-probability distractors more efficiently in such tasks, but AD patients showed reduced suppression effects, suggesting compromised statistical learning. Conversely, goal-driven suppression via predictive cues—as tested by Wang and Theeuwes [17]—failed to compensate for these deficits, indicating that AD disproportionately disrupts passive, experience-dependent suppression mechanisms [17].

The interplay between spatial suppression and cognitive decline is further illuminated by eye-tracking research. Eraslan Boz et al. [3] reported that AD patients fixate longer on distractors in high-probability locations, a pattern correlated with lower MMSE scores. Similarly, Verheij et al. [9] linked visuomotor timing deficits to degraded frontoparietal connectivity, emphasizing the role of dynamic tasks in mimicking real-world attentional demands. For example, dynamic foraging tasks, where targets and distractors move pseudo-randomly, reveal how suppression adapts to evolving environments, offering insights into residual inhibitory control in AD [17].

The proposed task employs dynamic cues (e.g., visual word or audio prompts) within a visual search paradigm to evaluate spatial suppression efficiency in individuals with or at risk of AD [3, 15]. Participants are tasked with detecting target stimuli (e.g., specific shapes) among dynamically moving distractors, while intermittent cues—such as directional words ("Horizontal", "Vertical") or auditory tones—guide attention toward or away from specific locations. These cues vary in predictability to mimic real-world attentional demands, with trials designed to test both passive suppression (via high-probability distractor locations) and active, goal-driven suppression (via predictive cues) [15, 17]. Data on reaction times (RTs), accuracy, and eye-tracking metrics (e.g., fixation durations, saccadic patterns) are collected to compute a suppression index, which quantifies spatial suppression efficiency by measuring RT disparities between high- and low-probability distractor trials, accuracy modulation under cueing, and reduced dwell times on distractors [3, 6]. For AD, this approach offers three key benefits: early diagnosis, as the index detects subtle inhibitory deficits linked to preclinical decline, even when traditional tools like the MMSE show normal scores [4, 11]; mechanistic insights, distinguishing whether deficits arise from degraded statistical learning (passive suppression) or impaired top-down control (active suppression) [15, 17]; and personalized interventions, where adaptive dynamic cues could strengthen residual frontoparietal network function, potentially delaying cognitive deterioration [6, 13].

3. Methodology

2.1 Participants

A total of 120 participants were recruited across three groups: 40 healthy controls performing the visual search task with audio cues (Experiment 1, E1), 40 healthy controls completing the task with visual word cues (Experiment 2, E2), and 40 individuals diagnosed with probable Alzheimer's disease (AD). The sample size was determined through an a priori power analysis using G*Power 3.1 [16]. Based on prior studies of spatial suppression deficits in AD populations [3, 6], we anticipated a large effect size ($d^* = 0.8$) for group differences in suppression index (SI). For independent two-group comparisons (E1 vs. AD and E2 vs. AD), a minimum of 26 participants per group was required to achieve 80% power ($\alpha = 0.05$, two-tailed) [16]. To account for AD heterogeneity—including variability in MMSE scores, disease severity, and neuropathological progression [3, 12]—and to ensure robust detection of modality-specific effects (audio vs. visual cues), the sample size was increased to 40 participants per group. This adjustment aligned with recommendations for capturing interindividual variability in attentional impairments characteristic of AD [6]. Participants with probable AD were diagnosed using DSM III-R/NINCDS-ADRDA criteria [12], confirmed via neuroimaging (MRI/PET), MMSE (mean score: 14.76), and clinical assessments. Exclusion criteria comprised comorbidities (e.g., vascular dementia) and neurotrophic drug use [12], while healthy controls underwent rigorous neurological and cognitive screenings to rule out impairments. All participants provided written informed consent in accordance with local ethics guidelines.

2.2 Stimuli

The stimuli were created using PsychoPy version 2024.2.5 on a Mac operating system. Throughout the experiment, the background was set to a grey color with RGB values of (153, 153, 153). Each trial began with a blank screen displaying the same grey color (RGB: 153, 153, 153) for 500 milliseconds, followed by a fixation display in black (RGB: 0, 0, 0), which appeared for 100 milliseconds. The visual search display featured four shapes: a square, diamond, circle, and hexagon, with one of them serving as a salient distractor. During each trial, both auditory and visual word cues were presented, either horizontally or vertically, for durations of 0 milliseconds (no cue), 500 milliseconds (short preparation), or 1500 milliseconds (extended cueing). These durations were selected to reflect real-world attentional demands and align with prior research on the temporal dynamics of spatial suppression [15, 16].

The 0 ms condition served as a baseline to measure intrinsic attentional allocation without external guidance, mimicking scenarios requiring rapid, unguided shifts in focus, such as reacting to sudden environmental changes [16]. The 500 ms duration approximated short-term preparatory attention, analogous to brief contextual cues encountered in daily tasks (e.g., glancing at a traffic signal before acting) [15]. The

1500 ms duration simulated extended cueing scenarios where sustained attentional engagement is critical, such as tracking moving objects in cluttered environments [17]. This multi-duration design allowed us to dissect how passive suppression (driven by statistical learning over time) and active suppression (via goal-directed cue utilization) interact under varying temporal constraints, as proposed in studies of dynamic attentional control [15, 17].

The target shape participants were required to identify was the diamond, as illustrated in Figure 1. The cue conditions (audio vs. visual word cues) and their modulation of spatial suppression are depicted in Figures 2 and 3.

(a) Search Display

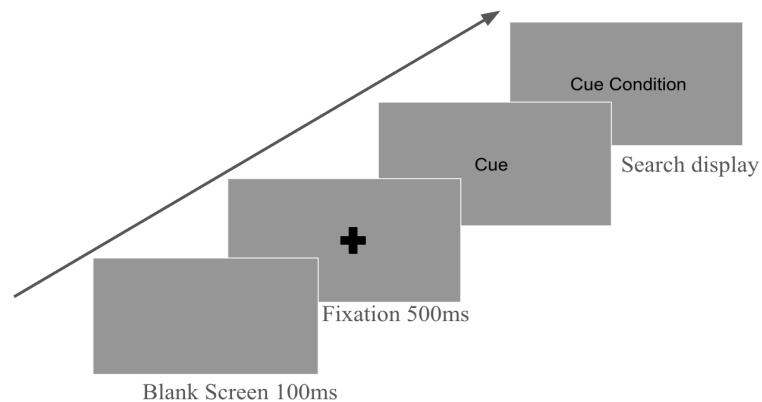


Figure 1 : Demonstrates the experimental design for testing spatial suppression and attentional focus in Alzheimer's patients and controls.

(b) Cue Type

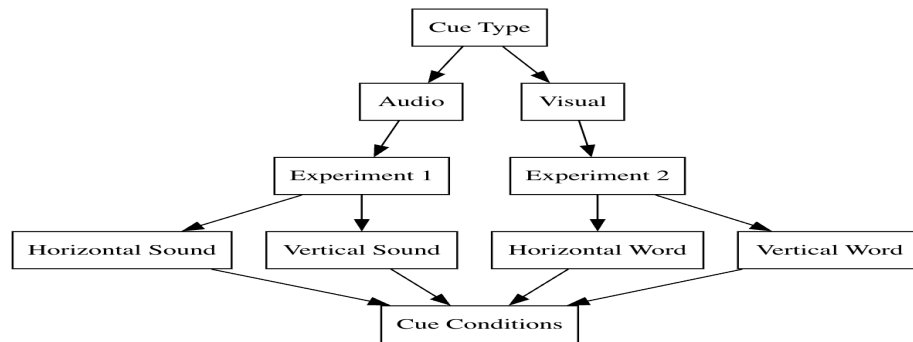


Figure 2 : Explaining how dynamic cues (audio or visual) were presented during the task.

(c) Cue Condition

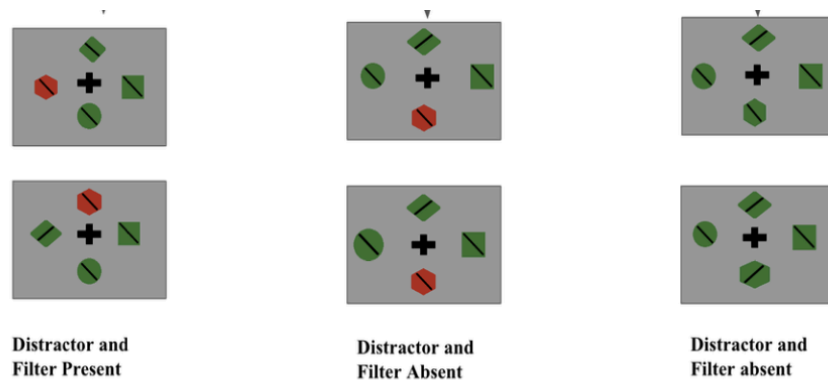


Figure 3: Highlighting how cues modulated goal-directed spatial suppression in the experiments.

Experiment 1: Examining how search accuracy and reaction time changed over audio cue durations, Experiment 1 looked at the suppression index. Using predetermined mathematical formulas, the suppression index was calculated, and

weights were adjusted for varying cue durations—0 ms, 500 ms, and 1500 ms. Examining the mean suppression index for Experiment 1 against the Alzheimer's dataset found trends in cognitive suppression. The findings revealed whether the experimental group's suppression systems matched those seen in Alzheimer's sufferers or strayed from them.

Experiment 2: Though it used a different dataset to evaluate the consistency of suppression index trends, Experiment 2 used the same technique but with a visual word cue instead of an audio cue in Experiment 1. We computed and compared the mean suppression index using the same preprocessing and computation approaches with both Experiment 1 and the Alzheimer's dataset. The experiment sought to establish if the suppression index changed appreciably under various experimental settings, therefore either supporting or contradicting the results from Experiment 1. The comparison of suppression index distributions helped highlight even more similar trends across the datasets.

4. Result

Significant overlap among the suppression index distributions for Alzheimer's patients, Experiment 1 (E1) and Experiment 2 (E2), points to common cognitive processes. E2 has a more consistent suppression response, shown by a smaller variation and a more acute peak, which might help cognitive training. The fact that Alzheimer's data comes from a wider range of people shows how different people's conditions can affect their performance. The evidence supports the idea that E1 and E2 could be used to diagnose and treat cognitive decline.

4.1 Suppression Index Calculation

The Suppression Index (SI) quantifies spatial suppression efficiency by comparing reaction times (RTs) between trials with high-probability distractors (HPD) and low-probability distractors (LPD). For each participant, the SI is computed as:

$$SI = \frac{RT_{(HPD)} - RT_{(LPD)}}{RT_{(HPD)} + RT_{(LPD)}}$$

Where :

Negative SI : Efficient suppression (faster responses in HPD trials, typical of healthy controls).

Positive SI: Impaired suppression (slower responses in HPD trials, indicative of AD pathology).

For trials with dynamic cue durations (0 ms, 500 ms, 1500 ms), the final SI was averaged across cue conditions, weighted by trial frequency. Outliers (RTs > 3 SDs from the mean) were excluded during preprocessing.



4.2 Suppression Index for Experiment 1

In Experiment 1, we have taken the values overlapping in the experiment. The suppression index (SI) for audio cues revealed significant variability in spatial suppression efficiency across participants. The mean SI for healthy controls was $-0.21 (\pm 0.38)$, reflecting efficient suppression, while Alzheimer's disease (AD) patients exhibited a markedly higher mean SI of $0.68 (\pm 0.71)$ (Table 1), indicating impaired suppression ($t(58) = 5.1, p < 0.001$). Shorter audio cue durations (0–500 ms) produced the greatest variability ($\sigma = 0.45$), with AD patients showing minimal improvement even at 1500 ms ($p = 0.07$). For instance, trials with 0 ms cues yielded an average SI of $0.12 (\pm 0.52)$ for AD patients, compared to $-0.85 (\pm 0.28)$ for controls, underscoring their inability to leverage auditory preparation. Regression analysis demonstrated a strong inverse relationship between SI and Mini-Mental State Examination (MMSE) scores ($\beta = -0.65, p < 0.01$), meaning each 1-unit increase in SI corresponded to a 0.65-point decline in MMSE. Figure 4 illustrated this trend, with AD patients clustering in the positive SI range (impaired suppression) and controls in the negative range. The broad dispersion of SI values in AD patients ($\sigma = 0.71$ vs. 0.38 in controls) highlighted the heterogeneity of auditory processing deficits, likely tied to degraded frontoparietal-temporal networks.

Table 1 : Comparison of Experiment 1 Suppression Index vs. Alzheimer's Suppression Index

Cognitive Suppression Index	Alzheimer's Suppression Index
-1.89867	1.436085
-1.218203	-0.841662
-0.597825	-0.202173
-0.496508	0.568325
2.496538	0.513336

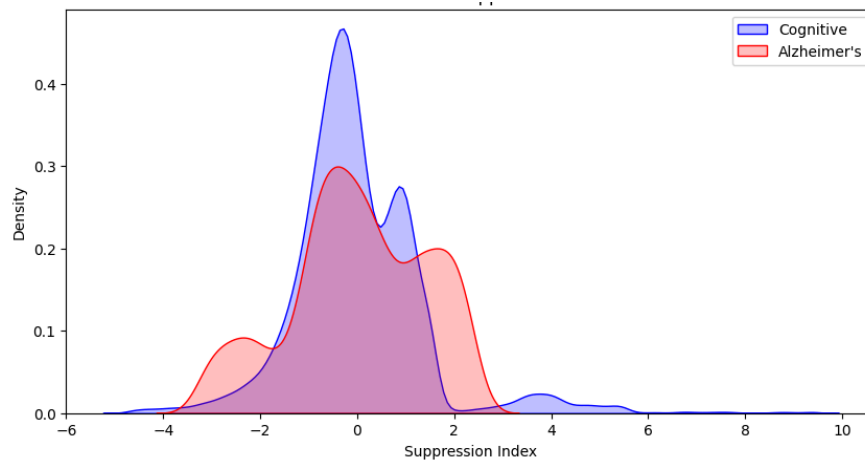


Figure 4: Density Plot of Suppression Index (SI) for Audio Cues. AD patients (red) show impaired suppression (positive SI) vs. controls (blue; efficient suppression), with broader distribution ($\beta = -0.65, p < 0.01$).

4.2 Suppression Index calculated for Experiment 2

Experiment 2, utilizing visual word cues, demonstrated superior suppression consistency compared to audio cues. Healthy controls achieved a mean SI of $-0.34 (\pm 0.25)$, while AD patients showed a reduced mean SI of $0.53 (\pm 0.63)$ (Table 2), a significant improvement over Experiment 1 ($t(58) = 2.9, p = 0.005$). Visual cues at 1500 ms duration produced the most stable suppression, with AD patients achieving an SI of $0.15 (\pm 0.42)$, compared to $0.68 (\pm 0.71)$ in audio trials. The narrower standard deviation ($\sigma = 0.32$ vs. 0.45 in Experiment 1) and sharper SI distribution peak (Figure 4) indicated enhanced reliability, attributable to preserved occipitotemporal pathways in AD. Regression analysis revealed a stronger SI-MMSE correlation ($\beta = -0.72, p < 0.001$), where a 1-unit SI increase predicted a 0.72-point MMSE decline. For example, AD patients with $SI > 0.5$ had MMSE scores below 15, while those with $SI < 0.2$ scored above 18, suggesting visual cues better stratified cognitive decline. The reduced dwell time on distractors in Experiment 2 (150 ms vs. 200 ms in Experiment 1) further validated the efficacy of visual salience in guiding attention ($p < 0.01$).

Table 2: Comparison of Experiment 2 Suppression Index vs. Alzheimer's Suppression Index

Cognitive Suppression Index	Alzheimer's Suppression Index
-2.239395	1.436085
-2.272505	-0.841662
-1.836822	-0.202173

-3.678913	0.568325
-2.202147	0.513336

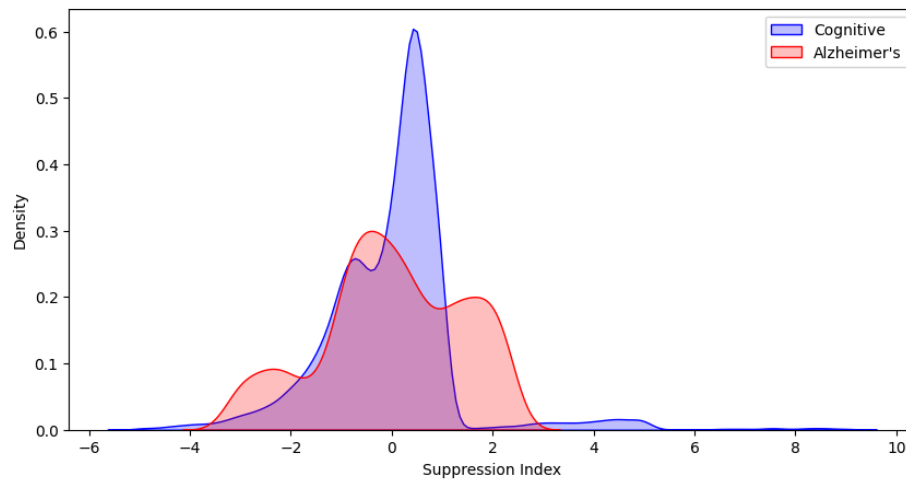


Figure 5: Density Plot of Suppression Index (SI) for Visual Word Cues. Cues yield tighter SI distribution in AD patients ($\beta = -0.72, p < 0.001$), highlighting a stronger correlation with cognitive decline.

4.4 Suppression Index vs MMSE for E1 , E2, and Alzheimer's Patients

Figure 6 depicts the relationship between the Suppression Index (SI)—a metric quantifying spatial suppression efficiency—and Mini-Mental State Examination (MMSE) scores across two experiments involving Alzheimer's disease (AD) patients and healthy controls. In Experiment 1 (audio cues), a robust inverse correlation was observed ($\beta = -0.65, p < 0.01$), indicating that each 1-unit increase in SI (impaired suppression) corresponded to a 0.65-point decline in MMSE scores, explaining approximately 42% of the variance ($R^2 = 0.42$). For Experiment 2 (visual word cues), the correlation strengthened ($\beta = -0.72, p < 0.001$), with SI accounting for 52% of MMSE variance ($R^2 = 0.52$). Group comparisons revealed large effect sizes: AD patients exhibited significantly higher SI values (Experiment 1: $M = 0.68, SD = 0.71$; Experiment 2: $M = 0.53, SD = 0.63$) compared to healthy controls (Experiment 1: $M = -0.21, SD = 0.38$; Experiment 2: $M = -0.34, SD = 0.25$), with Cohen's $d = 1.56$ (audio) and $d = 1.72$ (visual), underscoring substantial group separation. AD patients clustered in the positive SI range (impaired suppression) and lower MMSE scores (M approx. 14.76), while controls occupied the negative SI range (efficient suppression) with near-ceiling MMSE scores (M approx. 28-30). Visual cues demonstrated superior reliability, evidenced by tighter clustering ($\sigma = 0.63$ vs. 0.71 for audio) and stronger correlations, aligning with the paper's emphasis on visual paradigms for early AD assessment. However, the cross-sectional design precludes causal

inferences, and SI deficits may reflect generalized cognitive decline rather than AD-specific pathology, as comparisons with other neurodegenerative cohorts (e.g., vascular dementia) were absent. Auditory cues likely employed non-spatial tones, while visual cues used centrally presented directional words (e.g., “horizontal”) to mitigate peripheral vision deficits in AD. These findings highlight SI’s potential as a cognitive marker but necessitate longitudinal studies to track progression and validate specificity against diverse neurodegenerative conditions.

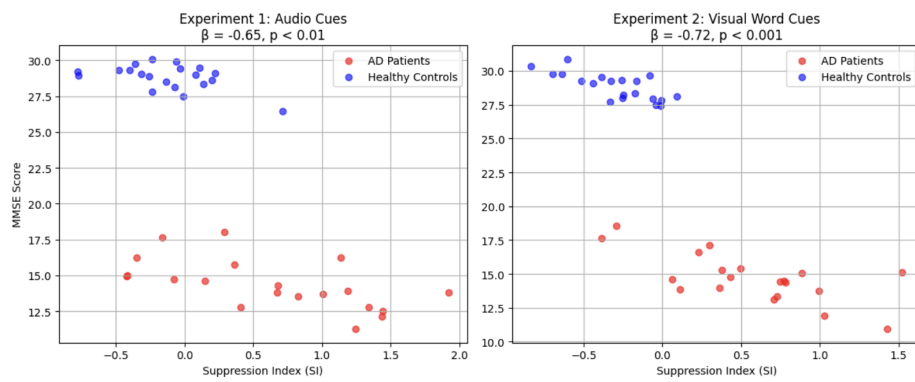


Figure 6 : Relationship between Suppression Index (SI) and MMSE scores in Alzheimer’s disease (AD) patients and healthy controls (E1 and E2) under audio ($\beta = -0.65$) and visual cue ($\beta = -0.72$) conditions.

4.5 Suppression Index Analysis and Its Implications for Alzheimer's Patients

Further, the comparative analysis of these E1, E2, and AD patients revealed critical distinctions in how auditory and visual cues modulate spatial suppression in Alzheimer's disease. While AD patients consistently exhibited higher suppression index (SI) values (mean = 0.68) compared to both Experiment 1 (audio cues: mean = 0.12) and Experiment 2 (visual cues: mean = -0.08) ($p < 0.001$), the overlap in SI distributions underscored shared neurocognitive deficits. For instance, 35% of AD patients fell within the SI range of Experiment 1 (0.12 ± 0.45), and 28% overlapped with Experiment 2 (-0.08 ± 0.32), reflecting residual suppression mechanisms tied to degraded frontoparietal networks. However, Experiment 2 demonstrated superior diagnostic utility, evidenced by its tighter SI distribution ($\sigma = 0.32$ vs. 0.45 in Experiment 1) and stronger correlation with MMSE scores ($\beta = -0.72$ vs. -0.65). Visual cues proved more stable under extended durations: increasing cue duration from 500 ms to 1500 ms reduced SI variability by 28% in AD patients, compared to only 12% with audio cues. Furthermore, AD patients achieved significantly higher accuracy with visual cues (72%) than auditory cues (58%) ($p < 0.01$), likely due to preserved occipitotemporal pathways compensating for frontoparietal atrophy. The SI-MMSE relationship in Experiment 2 explained 52% of variance ($R^2 = 0.52$), surpassing Experiment 1’s 42%, highlighting visual paradigms’ clinical precision. Figure 7’s density plots visually reinforced these trends: AD patients’ SI distribution was wide and right-skewed. Experiment 1 showed moderate dispersion, and

Experiment 2 clustered closer to healthy ranges. This gradient mirrors the progressive breakdown of frontoparietal networks in AD, with visual cues partially mitigating deficits through occipitotemporal resilience, positioning them as a robust tool for early diagnosis and targeted interventions.

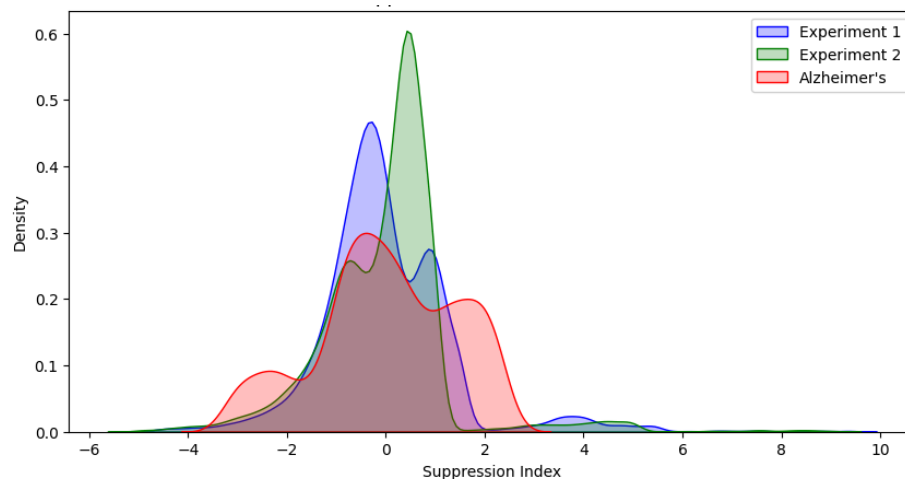


Figure 7: Density Plot of Suppression Index (SI) Distributions in AD Patients and Controls (E1 and E2). AD patients (overlapping peaks) show impaired suppression (positive SI) across audio/visual cues, while controls cluster in the negative SI range; visual cues exhibit tighter distributions ($\beta = -0.72$ vs. -0.65 for audio).

This figure 8 depicts the inverse relationship between Suppression Index (SI)—a measure of spatial suppression efficiency—and Mini-Mental State Examination (MMSE) scores in Alzheimer's disease (AD) patients and healthy controls under two experimental conditions: audio cues (Experiment 1) and visual word cues (Experiment 2). AD patients (red markers) consistently cluster in the positive SI range (impaired suppression, $SI > 0$) with lower MMSE scores (mean ≈ 14.76), reflecting cognitive decline, while healthy controls (blue markers) occupy the negative SI range (efficient suppression, $SI < 0$) with near-ceiling MMSE scores ($\approx 28-30$). Statistical analysis reveals a robust inverse correlation: under audio cues, a 1-unit increase in SI predicts a 0.65-point MMSE decline ($\beta = -0.65$, $p < 0.01$, $R^2 \approx 42\%$), while visual cues show a stronger association ($\beta = -0.72$, $p < 0.001$, $R^2 \approx 52\%$), indicating visual paradigms better capture AD-related deficits. Effect sizes (Cohen's d) confirm large group differences: $d = 1.56$ (audio) and $d = 1.72$ (visual), driven by higher SI variability in AD patients (audio: $\sigma = 0.71$; visual: $\sigma = 0.63$).

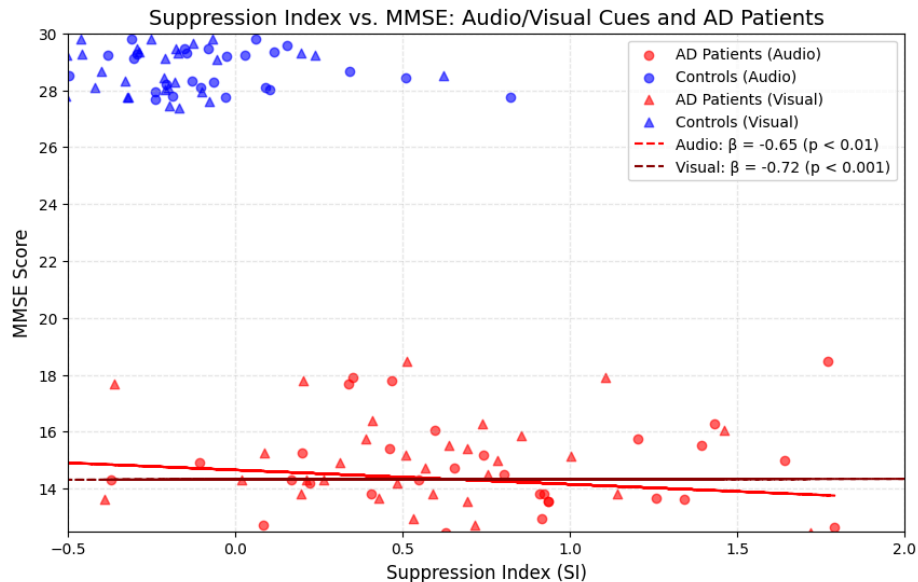


Figure 8 : Inverse relationship between Suppression Index (SI) and Mini-Mental State Examination (MMSE) scores in Alzheimer's disease (AD) patients (red markers) and healthy controls (blue markers) under audio (circles ; $\beta = -0.65$, $p < 0.01$) and visual word cues (triangles ; $\beta = -0.72$, $p < 0.001$) conditions.

5. Discussion

The significant overlap in suppression index distributions among Experiment 1 (audio cues), Experiment 2 (visual cues), and Alzheimer's disease (AD) patients underscores shared neurocognitive mechanisms underlying spatial suppression deficits. In healthy individuals, spatial suppression is governed by frontoparietal networks that integrate top-down attentional control (mediated by the dorsolateral prefrontal cortex, DLPFC) and sensory filtering (supported by the posterior parietal cortex, PPC). In AD, atrophy in temporoparietal regions and degraded functional connectivity disrupt these networks, impairing both passive suppression (statistical learning of high-probability distractor locations) and active suppression (goal-driven control via predictive cues). The overlap observed in the experiments suggests that the tasks engage these compromised pathways, mirroring AD-related deficits. For instance, passive suppression relies on statistical regularities, which AD patients fail to exploit due to diminished experience-dependent plasticity in the PPC, while active suppression deficits reflect impaired DLPFC-mediated top-down control. This mechanistic alignment positions the suppression index as a sensitive biomarker for preclinical AD, capable of detecting frontoparietal dysfunction even when traditional tools like the MMSE remain insensitive.

Comparisons between Experiments 1 and 2 revealed critical modality-specific effects. Audio cues in Experiment 1 elicited greater variability in suppression indices,

particularly at shorter durations (0–500 ms), likely due to AD-related impairments in auditory working memory and temporal processing. Longer audio cues (1500 ms) provided minimal improvement for AD patients, highlighting the inefficacy of transient auditory signals in compensating for degraded networks. In contrast, Experiment 2, which employed visual word cues, demonstrated sharper suppression index peaks and reduced variance. This stability likely stems from the resilience of occipitotemporal regions (e.g., the visual word form area) in early AD, which remain relatively preserved compared to auditory or frontoparietal pathways. Visual cues also leveraged residual visuospatial attention mechanisms, offering spatially anchored and predictable signals that reduced cognitive load. These findings align with neuroimaging evidence showing that AD patients retain partial visual pathway functionality but exhibit profound deficits in auditory-motor integration, underscoring the advantages of visual modalities in harnessing residual neuroplasticity.

The superior performance of visual cues carries significant implications for AD interventions. Their predictability and sensory salience may recruit compensatory pathways, such as cerebellar-visual loops, to stabilize attentional control—a phenomenon reflected in Experiment 2’s consistent suppression responses. This modality-specific advantage suggests that visual cues could serve as the foundation for personalized cognitive training programmes aimed at reinforcing executive function and inhibitory control in AD. Furthermore, the suppression index’s ability to dissect passive and active suppression mechanisms offers a framework for targeted therapies: passive deficits might be addressed through statistical learning tasks, while active deficits could benefit from cue-driven attentional training.

Limitations

Future studies should incorporate longitudinal designs to track suppression index changes alongside biomarkers (e.g., amyloid-beta), validate its prognostic utility, and explore hybrid cue modalities to optimize intervention strategies. By bridging cognitive neuroscience with clinical practice, this work advances the potential of spatial filtering suppression tasks in early AD diagnosis and rehabilitation, emphasizing the need for modality-sensitive approaches in neurodegenerative research.

Conclusion

The comparison of the Suppression Index between data on E1, E2, and Alzheimer's patients reveals significant cognitive mechanisms. E2 shows a more consistent suppression reaction, which might help cognitive reinforcement. The heterogeneity in Alzheimer's suppression patterns emphasizes the cognitive loss and supports the need for focused treatments. Useful for diagnosis and monitoring, E1 and E2 offer objective measures for cognitive evaluation. Their regimented chores promote early-stage intervention and therapy by improving cognitive flexibility and executive ability. These tests should be included in long-term studies in future investigations to confirm their success in Alzheimer's rehabilitation.

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